

Chapter 19 — Judgement

When not to use any of this

Eighteen chapters of capability. This one is about the four places you should put it down.

The slide to hold in the head

- **Not for a novel derivation.** If the step has never been written down, there is nothing to predict from. You are asking a next-token predictor to be first.
- **Not for anything you cannot verify.** If checking the answer is harder than producing it, you have moved work, not saved it.
- **Not where the failure would be silent.** A wrong sign that still runs, a plausible coefficient, a citation that resolves to a real paper saying something else.
- **Not with data that cannot leave the building.** Unpublished results, sponsor-confidential geometry, anything under an NDA.

Four tests. If a task fails any one of them, the tool is the wrong instrument — not a risky instrument, the wrong one.

Why these four and not a longer list

- **Novel derivation** follows from Chapter 1. The objective is next-token prediction over what has been written. Genuinely new mathematics is the case with no support in the distribution.
- **Cannot verify** follows from Chapter 4. Without a check you cannot tell an improvement from a coincidence, and Chapter 5 says fluent output is not evidence of correctness.
- **Silent failure** follows from the failure gallery. Every item in it was caught by something. The dangerous class is the one with nothing to catch it.
- **Data that cannot leave** follows from Chapters 12 and 18. You opened those channels deliberately; this is the boundary you set.

A longer list would be forgotten by Thursday. These four are derivable from the course, which is why they will survive it.

What has not changed

- The claim is yours. The interpretation is yours. The responsibility for correctness is yours.
- None of those is delegable, and no disclosure statement makes them so.
- A tool that drafts faster raises the volume you are accountable for. That is the trade you have actually made this term.

Capstone

One end-to-end task on your own work, using what you configured across the three sessions.

- Your project instruction file from Chapter 10.
- Your eval worksheet from Chapter 4, on a claim you are relying on.
- Your verification habit from Chapter 14, on every source it hands you.
- Your disclosure position from Chapter 18, written down before you need it.

If a piece of that stack is missing, that is the honest finding of the day, and it is more useful than a finished artefact.

Round-robin

One thing each person will use this week.

- Not the most impressive thing. The one you will actually do before Friday.
- Say it out loud. Committing in front of five colleagues is the only enforcement mechanism this course has.

What you take away

- The repository — slides, run sheets, the eval worksheet, the failure gallery.
- The one-page reference card.
- A shared instruction file the group owns, not six private ones.

And the habit the whole course was built around: **state the check before you do the work.**

Everything else here has a shelf life. That does not.